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RobSense: A Robust Multi-modal Foundation Model for Remote Sensing with Static, Temporal, and Incomplete Data Adaptability

Minh Kha Do, Kang Han, Phu Lai, Khoa T. Phan, and Wei Xiang*
La Trobe University
{m.do, k.han, p.lai, k.phan, w.xiang}@latrobe.edu.au

Abstract

Foundation models for remote sensing have garnered increasing attention for their strong performance across various observation tasks. However, current models lack robustness in managing diverse input types and handling incomplete data in downstream tasks. In this paper, we propose RobSense, a robust multi-modal foundation model for Multi-spectral and Synthetic Aperture Radar data. RobSense is designed with modular components and pre-trained by a combination of temporal multi-modal alignment and masked autoencoder strategies on a huge-scale dataset. Therefore, it can effectively support diverse input types, from static to temporal, uni-modal to multi-modal. To further handle the incomplete data, we incorporate two uni-modal latent reconstructors that recover rich representations from incomplete inputs, addressing variability in spectral bands and temporal sequence irregularities. Extensive experiments demonstrate that RobSense consistently outperforms state-of-the-art baselines on complete datasets across four input types for segmentation, classification, and change detection. On incomplete datasets, RobSense outperforms the baselines by considerably larger margins when the missing rate increases. Project page: https://ikhado.github.io/robsense/

1. Introduction

Remote sensing data play a crucial role in Earth observation, offering a rich array of information essential for understanding our planet. To analyze this data effectively across tasks such as land use and land cover mapping, scene classification, and more, deep learning models have become increasingly popular [23]. Typically, these models rely on supervised learning, which requires a substantial amount of annotated data to achieve high performance on specific tasks. However, collecting and labeling this data demands significant resources and considerable human effort [10], posing a major challenge. This challenge is especially pronounced in remote sensing, where annotation requires specialized expertise to accurately interpret the data [27].

Meanwhile, there is a large volume of unlabelled data collected from Sentinel missions. These missions mainly consist of Sentinel-2 acquiring Multi-spectral (MS) imagery and Sentinel-1 acquiring Synthetic Aperture Radar (SAR) data. MS imagery provides rich spatial information and enables the identification and analysis of the material composition of objects across multiple spectral bands [8]. SAR data offers all-weather, day-and-night radar, allowing complete monitoring of surface changes regardless of weather conditions [11]. Therefore, designing multi-modal foundation models that integrate both data types through self-supervised pre-training is crucial [7, 8, 21], as it significantly reduces the amount of annotated data required for downstream tasks.

Foundation models have achieved remarkable success in diverse domains, including computer vision and language processing [13, 26]. However, their development for remote sensing remains underexplored. Recently, emerging models have begun to address this gap [4, 7]. These models can be classified as either uni-modal, which handles a single type of data, or multi-modal, which integrates multiple data types. Both typically employ a self-supervised learning approach. This approach leverages Masked Autoencoder strategy (MAE) [4, 20], which involves reconstructing masked portions of the input. For uni-modal models, the typical input consists of MS images. These images can be divided into patches across all bands [4] or 3D tube-shaped patches [9] spanning several adjacent bands. Meanwhile, multi-modal models integrate SAR data with MS images. To capture the latent representations between these two modalities, contrastive learning is commonly applied alongside MAE [7]. While these models demonstrate promising results on various downstream tasks, two key robustness challenges remain: supporting diverse input types and effectively handling incomplete data.

*Corresponding author.

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The first challenge is designing a foundation model that supports diverse input types in downstream tasks. The current models only support limited input types. Tab. 1 illustrates the adaptability of recent representative models across four input types. SatMAE++ [18] mainly supports static MS inputs. While SatlasNet [1] and SpectralGPT [9] can process from static to temporal MS inputs, they are not capable of handling multi-modal input. In contrast, Croma [7] processes multi-modal data by integrating MS and SAR data with patch contrastive learning, applying contrastive loss to spatially aligned MS-SAR patch pairs as positives and non-aligned patches as negatives. However, it heavily relies on data augmentation, which can lead to information loss in latent representations from two heterogeneous modalities. This issue is especially evident when MS-SAR patches are misaligned or neighboring patches representing the same scene are mistakenly treated as negative pairs across modalities [7, 30]. Moreover, Croma is limited to static inputs and does not support temporal downstream datasets.

The second challenge is to enhance model performance when handling incomplete downstream datasets. None of the existing models directly address the incomplete data in the downstream tasks. Two critical scenarios often cause this incompleteness: (i) varying the number of spectral bands and (ii) irregularities in temporal sequences. Firstly, existing models typically require downstream datasets to have a fixed number of spectral bands as in the pre-training datasets. However, the number of spectral bands can vary due to different capturing instruments in practice, *e.g*. Sentinel-2, Landsat-8, *etc*., [15] as well as factors such as extreme atmospheric conditions or physical degradation [17]. Therefore, this variation limits the adaptability of models in downstream datasets affected by such conditions. Secondly, in practice, capturing data frequently from the same location is challenging [15]. This infrequency leads to time point gaps in temporal sequences. These gaps lead to missing information about fluctuations in human activities or seasonal patterns. This issue could result in sparse representations in the latent space, causing information loss while fine-tuning models.

To address the aforementioned challenges, we propose a robust multi-modal foundation model tailored for MS and SAR data, namely RobSense. We design two parallel Vision Transformer [6] (ViT)-based encoders for uni-modal inputs, called uni-modal encoders, along with a multi-modal encoder that integrates representations from these uni-modal encoders to produce multi-modal representations. To facilitate multi-modal learning and prevent information loss, we employ two pre-training phases, namely temporal multi-modal learning and latent reconstruction learning. The first phase focuses on enhancing latent representations between two modalities. In order to preserve specific information about seasonal changes, human activities, *etc*., in the representations for each time-point, we employ multivariate Kullback-Leibler (KL) divergence as temporal distribution alignment (TDA). This technique treats each time step of the MS-SAR sequences as a Gaussian distribution, constraining two uni-modal representations at the same time point to the same distribution. The second phase aims to bolster the performance with incomplete inputs. Specifically, we design two latent reconstructors following two uni-modal encoders to recover rich representations caused by variational spectral bands or gaps in temporal sequences. In summary, the main contributions of this paper are as follows:

- We propose RobSense, a robust multi-modal foundation model for remote sensing that supports diverse input types, including static to temporal, uni-modal to multi-modal, and incomplete data.
- We propose a temporal multi-modal learning for remote sensing, which leverages temporal multi-modal data through distribution alignment over time to preserve time-specific distributions. Therefore, our model produces consistent and rich representations.
- To address the data incompleteness, we design two uni-modal latent reconstructors, enabling the recovery of rich representations from sparse ones caused by incomplete remote sensing data.
- Extensive experiments show that RobSense outperforms recent baselines on three key downstream tasks with the complete datasets. Furthermore, RobSense consistently surpasses baselines on incomplete downstream datasets across various missing rates, with a larger performance margin as the missing rate increases.

Table 1. Supported input types in recent representative remote sensing foundation models.

Model	Supported input type			
	Static uni-modal	Static multi-modal	Temporal uni-modal	Temporal multi-modal
SatlasNet [1]	✓		✓	
SpectralGPT [9]	✓		✓	
SatMAE++ [18]	✓			
Croma [7]	✓	✓		
RobSense (ours)	✓	✓	✓	✓

2. Related Work

Remote sensing foundation models have gained significant attention due to their ability to apply to various tasks. There are two primary approaches for training these models: multi-task learning and self-supervised learning.

Multi-task learning offers generalization benefits by leveraging shared information across related tasks. Wang *et al*. [25] adopted an aerial scene recognition dataset for pre-training three backbones, *i.e*., CNN, ViT, and ViTAE.

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These models were subsequently evaluated across four downstream tasks to assess their performance. SatlasNet [1] was simultaneously pre-trained on seven label types with seven heads, leveraged by the incorporation of different common tasks. Although these models demonstrate the ability to handle diverse downstream tasks, they require a large amount of annotated data, making them resource-intensive and challenging to scale.

Meanwhile, self-supervised learning has recently become more prevalent and provided an alternative by reducing dependency on annotated datasets [23]. A common strategy in self-supervised learning is MAE, which reconstructs the original input from randomly masked data portions. This strategy allows models to learn rich, generalized representations. SatMAE [4] is one of the pioneering models applying MAE to MS images, employing different masking strategies across spatial and temporal dimensions. Building on this, SatMAE++ [18] improves performance by utilizing multi-scale reconstruction with multi-level convolutional-based blocks. Danfeng *et al*. [9] introduced a 3D spatial-spectral masking strategy to preserve spectral continuity. While these models have demonstrated promising results on various downstream tasks, they are limited to the MS modality. Other efforts [7, 12, 16] have explored multi-modal data integration by combining MAE with a contrastive learning objective. This approach faces challenges with information loss, particularly when positive pairs in the contrastive learning objective are not well-aligned spatially. Moreover, these models predominantly process static inputs and lack robustness in the temporal dimension.

Unlike previous models, which are pre-trained on either static multi-modal data or static/temporal uni-modal data, the proposed model leverages temporal multi-modal data to capture rich representations across both multi-view and temporal dimensions. Therefore, it supports diverse input types and effectively addresses the critical issue of incomplete downstream datasets.

3. Method

In this section, we first detail the pre-training stage of RobSense, detailing each phase. We then demonstrate its downstream usage.

3.1. Pre-training

Fig. 1 illustrates the two phases of the pre-training stage: temporal multi-modal learning and latent reconstruction learning. The first phase focuses on training two uni-modal encoders, the multi-modal encoder, and the multi-modal decoder. This phase employs temporal distribution alignment to enable the encoders to generate rich representations from complete data. In the second phase, we train two uni-modal latent reconstructors to recover the rich representations from sparse ones caused by incomplete data. This phase ensures that the model is resilient to missing information. For both phases, the input is the temporal multi-modal pair, $pair = (ms, sar)$, with sequence length T.

3.1.1 Temporal Multi-modal Learning

The primary objective of this training phase is to train two uni-modal encoders (G_{MS} and G_{SAR} for MS and SAR, respectively), multi-modal encoder (MG), and multi-modal decoder (MD). The uni-modal encoders process the sequence of uni-modal time point $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ from the temporal pair (ms, sar), where H, W, and C denote the height, width, and number of bands, respectively. The time point $\mathbf{x}$ is patchified into non-overlapping patches as the sequence $\mathbf{x} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_L\}$, where $\mathbf{x}_i \in \mathbb{R}^{p\times p\times C}$, p is the patch size, $L = (H/p)\cdot(W/p)$ is the number of patches. As a result, the temporal dimension of ms and sar is (T, L, p, p, C). Then, each patch is passed into the patch embedding $f_p : \mathbb{R}^{p\times p\times C} \mapsto \mathbb{R}^d$ to create sequence (T, L, d) with embedding vector $\mathbf{x}' \in \mathbb{R}^{L\times d}$ of each time point $\mathbf{x}$. In order to enable efficient learning of rich representations, we employ the MAE [24] strategy, which randomly keeps a set $\mathbb{V}$ of visible patch embeddings and masks other invisible patch embeddings from each embedding vector $\mathbf{x}'$. We apply consistent masking [21], which means that the masking regions are consistent across all time points $\mathbf{x}$ from (ms, sar). The masking ratio is chosen as 90% to facilitate effective training [9]. The resulting masked embedding vector is denoted as $\tilde{\mathbf{x}}' \in \mathbb{R}^{|\mathbb{V}|\times d}$. To incorporate temporal information from the temporal sequences, we employ the joined embedding of position and acquisition day of the year (DOY) [31] corresponding to $\mathbf{x}$. The kth element of the embedding vector $PE(DOY_x)$ is defined as Eq. (1).

$$PE(DOY_{\mathbf{x}})_k = \begin{cases} \sin\left(\frac{DOY_i}{10000^{2j/d}}\right), & \text{if } k = 2j, \\ \cos\left(\frac{DOY_i}{10000^{2j/d}}\right), & \text{if } k = 2j+1. \end{cases} \quad (1)$$

The masked embedding vector with $PE(DOY_{\mathbf{x}})$ is passed into the uni-modal encoder. The time point representation of the time point $\mathbf{x}$ is computed as $z = G(\tilde{x}' + PE(DOY_{\mathbf{x}}))$. Therefore, the latent representations for ms and sar are presented as $\mathbf{z}_{ms}$ and $\mathbf{z}_{sar}$, respectively. After passing the uni-modal encoders, the zero vectors are inserted into the output to form the complete set of latent representations $(\mathbf{z}_{ms}, \mathbf{z}_{sar}) = \left(\{z^t_{ms}\}^T_{t=1}, \{z^t_{sar}\}^T_{t=1}\right)$.

Temporal Distribution Alignment Loss (TDA). In multi-modal learning, effectively extracting the information from the heterogeneous between modalities is important.

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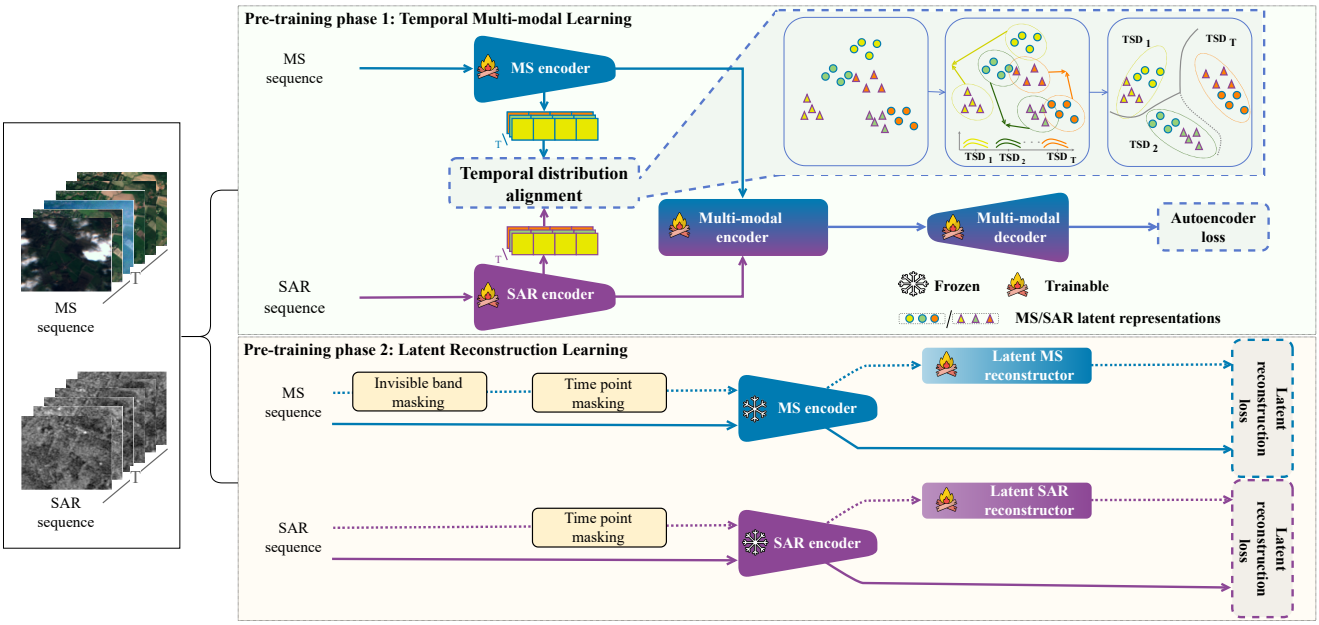


Figure 1. Illustration of RobSense pre-training, which consists of two training phases: *Temporal Multi-modal Learning* for training MS/SAR encoder, multi-modal encoder and multi-modal decoder, and *Latent Reconstruction Learning* for training Latent MS/SAR reconstructor. TSD stands for Time-specific Distribution.

The prior study [7] uses contrastive learning, which struggles to control positive and negative patch pairs, often causing information loss [4, 30]. To address this, we introduce temporal distribution alignment that leverages temporal multi-modal data to constrain latent representations from both modalities. This approach ensures that two uni-modal representations with similar timestamps follow a consistent distribution, preserving time-specific distribution at each time point. This enables the model to produce consistent, rich representations across both modal and temporal dimensions.

Every time point representation z within the temporal latent representations $(\mathbf{z}_{ms}, \mathbf{z}_{sar})$ is presented under a single Gaussian distribution $\mathcal{N}(\mathbf{x}; \mu, \sigma^2)$ with the mean μ and the standard deviation σ. Therefore, the whole temporal sequence is the multivariate Gaussian distribution. In particular, the modalities MS and SAR follow the continuous distributions $P_{ms}(\mathbf{z}) \sim \mathcal{N}(\mathbf{z}_{ms}; \boldsymbol{\mu}_{ms}, \boldsymbol{\Sigma}^2_{ms})$ and $P_{sar}(\mathbf{z}) \sim \mathcal{N}(\mathbf{z}_{sar}; \boldsymbol{\mu}_{sar}, \boldsymbol{\Sigma}^2_{sar})$ with the expectations $\boldsymbol{\mu} \in \mathbb{R}^{T\times 1}$ and the diagonal matrix of standard deviation $\boldsymbol{\Sigma} \in \mathbb{R}^{T\times T}$. KL divergence aligns the distributions of two temporal sequences by quantifying the divergence between their corresponding multivariate Gaussian distributions. In particular, the KL divergence from $P_{ms}(\mathbf{z})$ distribution to $P_{sar}(\mathbf{z})$ is presented as Eq. (2),

$$\mathrm{KL}_{(ms,sar)} = \int P_{ms}(\mathbf{z}) \log \frac{P_{ms}(\mathbf{z})}{P_{sar}(\mathbf{z})}\, d\mathbf{z}, \tag{2}$$

and the KL divergence can be expressed by Gaussian process as Eq. (3),

$$\mathrm{KL}_{(ms,sar)} = \frac{1}{2}\Bigg[\log\frac{|\boldsymbol{\Sigma}_{sar}|}{|\boldsymbol{\Sigma}_{ms}|} + \mathrm{Tr}(\boldsymbol{\Sigma}^{-1}_{sar}\boldsymbol{\Sigma}_{ms}) + \Delta^{\top}_{ms,sar}\boldsymbol{\Sigma}^{-1}_{sar}\Delta_{ms,sar} - T\Bigg], \tag{3}$$

where $\Delta_{ms,sar} = (\boldsymbol{\mu}_{sar} - \boldsymbol{\mu}_{ms})$. Since KL divergence is not symmetric, the TDA loss for this phase is the mean between $\mathrm{KL}_{(ms,sar)}$ and $\mathrm{KL}_{(sar,ms)}$ as Eq. (4),

$$\mathcal{L}_{\mathrm{TDA}} = \frac{1}{2}\left(\mathrm{KL}_{(ms,sar)} + \mathrm{KL}_{(sar,ms)}\right). \tag{4}$$

Autoencoder Loss. The temporal latent representations pair $\mathbf{z}_{ms}, \mathbf{z}_{sar}$ are concatenated and forwarded through the multi-modal encoder and decoder to recover the original patches. The set of recovered patches for each time point $\mathbf{x}$ is denoted as $\hat{\mathbf{x}}$. The autoencoder loss ($\mathcal{L}_{\mathrm{AE}}$) is computed as Eq. (5),

$$\mathcal{L}_{\mathrm{AE}} = \frac{1}{N_{\mathbb{V}}}\sum_{t=1}^{T}\sum_{j\notin\mathbb{V}}\Big[\left(\mathbf{x}^t_{j,MS} - \hat{\mathbf{x}}^t_{j,MS}\right)^2 + \left(\mathbf{x}^t_{j,SAR} - \hat{\mathbf{x}}^t_{j,SAR}\right)^2\Big], \tag{5}$$

where $N_{\mathbb{V}} = 2T(L - |\mathbb{V}|)$ is the total number of invisible patches. The objective is to train the encoders to learn rich

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representation by recovering the invisible patches. In the first pre-training phase, the objective function is formulated by combining the TDA loss and the Autoencoder loss, as $\mathcal{L} = \mathcal{L}_{\text{TDA}} + \mathcal{L}_{\text{AE}}$.

3.1.2 Latent Reconstruction Learning

This phase aims to enhance the performance of the model to data incompleteness. As shown in Fig. 1, two uni-modal latent reconstructors, *i.e.*, latent MS reconstructor (R_{MS}) and latent SAR reconstructor (R_{SAR}) are trainable, and other modules are frozen.

Satellite-based remote sensing data acquisition is often incomplete due to various operational and environmental factors [15]. Furthermore, the diversity of sensors leads to variations in the spectral bands. Consequently, there are two main scenarios: (i) variability in the number of spectral bands and (ii) irregularities in the temporal sequences. Therefore, this phase trains the model to be more resilient, creating two input scenarios by randomly removing (i) invisible spectral bands from MS images and (ii) time points from MS and SAR sequences.

Band masking in MS images. This scenario simulates the potential absence of certain spectral bands in MS image that may occur in real-world situations. To simulate this, some spectral bands are randomly removed, and zero-valued bands are padded to maintain the total number of bands as C. Let $\mathbf{x} = \{x^{(1)}, x^{(2)}, \ldots, x^{(C)}\}$ be a MS image with C spectral bands. The first three bands (i.e., RGB) are not often absent in remote sensing acquisitions and provide valuable information for reconstructing other spectral bands [2, 14]. Therefore, these bands are kept, while the remaining invisible bands are randomly removed. The resultant image $\mathbf{x}_{\text{drop}}$ is formed by padding the retained channels with a sequence of zero channels, ensuring the total number of channels is C as shown in Eq. (6),

$$\mathbf{x}_{\text{drop}} = \{x^{(1)}, x^{(2)}, x^{(3)}\} \cup \{x^{(c)} \mid c \notin \mathcal{C}_{\text{drop}}\} \cup \{0^c\}^{|\mathcal{C}_{\text{drop}}|}, \tag{6}$$

where $\mathcal{C}_{\text{drop}}$ is the set of indices corresponding to the dropped channels, and $0^c \in \mathcal{R}^{H\times W\times 1}$ is the zeros channel. All the images from the temporal sequence ms are consistently removed with the same band indices to form the sequence ms_{drop}.

Time point masking in sequences. The second scenario focuses on the temporal sequences, where time points from both temporal sequences of ms_{drop} and sar are randomly omitted during training. The approach is similar to the band removal in MS image. A temporal sequence is removed with some time points $\mathbf{x}$ and padded with zero values to keep the total length as T. As a result, the final incomplete pair is presented as $\tilde{pair} = (\tilde{ms}_{\text{drop}}, \tilde{sar})$.

Latent Reconstruction Loss. The main purpose of this loss is to minimize the distance between the reconstructed and the rich representations. Mathematically, the latent reconstruction loss $\mathcal{L}_{\text{LatRes}}$ is formulated as Eq. (7) with MSE is the Mean Square Error function,

$$\begin{aligned}\mathcal{L}_{\text{LatRes}} = &\text{MSE}\Big(G_{MS}(ms), R_{MS}\big(G_{MS}(\tilde{ms}_{\text{drop}})\big)\Big) \\ &+ \text{MSE}\Big(G_{SAR}(sar), R_{SAR}\big(G_{SAR}(\tilde{sar})\big)\Big).\end{aligned} \tag{7}$$

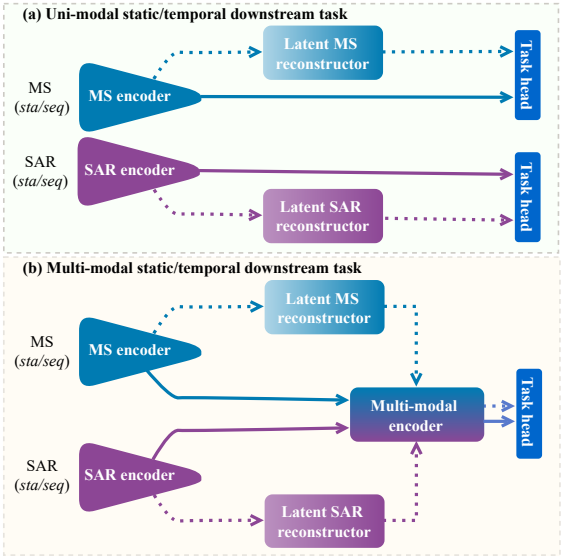


Figure 2. Downstream usage of RobSense. By combining different modules, the model can accept either static (*sta*) or sequential (*seq*) inputs. It supports both complete (solid line) and incomplete (dashed line) inputs.

3.2. Downstream Usage

Fig. 2 illustrates different modules that can be flexibly combined to support diverse input types, from uni-modal to multi-modal, from static to temporal, and incomplete data. Specifically, the uni-modal encoders can process uni-modal inputs independently (Fig. 2a) or work alongside the multi-modal encoder to accommodate multi-modal inputs (Fig. 2b). These uni-modal encoders could be followed by the uni-modal latent reconstructors to recover the rich representations from incomplete inputs.

4. Experiments

4.1. Setting

Pre-training. RobSense is pre-trained on SatlasPretrain dataset [1], which covers all continents except Antarctica. This huge-scale dataset provides both spatial and temporal contexts for multi-modal data, including 9-band MS imagery (B02–B08, B11, and B12) and 2-band SAR data

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(VH and VV). Following Favyen et al. [1], each sequence is limited to $T = 8$ time points, sorted by acquisition date. Temporal alignment is maintained by constraining the time gap between each pair ms_i and sar_i to a maximum of 30 days. A ViT-Base backbone with 12 layers is used for both the uni-modal and multi-modal encoders. Each uni-modal latent reconstructor has 4 ViT layers due to low-dimensional representations. The decoder, typically narrower than the encoder [9], has half as many layers, consisting of 6 ViT layers. The proposed model is pre-trained in two phases, processing temporal multi-modal MS-SAR data pairs. It operates at a spatial resolution $(224, 224)$ with a patch size of 16. Each phase uses a batch size of 8, an initial learning rate of $1e-4$, and a half-cycle cosine decay over 30 epochs. Pre-training utilizes 8 NVIDIA A100 80GB GPUs, totaling about 4,500 GPU hours.

Fine-tuning. We quantitatively assess the performance of the foundation models on three downstream tasks, *i.e.*, semantic segmentation, classification, and change detection. For comparison, we employ recent models with publicly available pre-trained weights, as shown in Tab. 1. All models are fine-tuned by a single GPU.

Semantic Segmentation. We use the labelled dataset from SatlasPretrain [1] for land use/land cover segmentation, containing around 53K temporal sequence pairs with MS and SAR modalities. The data is split into 80% for training and 20% for evaluation. Following Danfeng *et al*. [9], we utilize UperNet [29] as a segmentation head and apply a soft target cross-entropy loss function. We evaluate the fine-tuned models using the mean Intersection over Union (mIoU) metric across 11 categories.

Classification. We employ BigEarthNet dataset [19], which consists of static pairs of MS and SAR modalities. The MS modality has 12 bands, though some are non-informative [4]; our model utilizes nine, consistent with the SatlasPretrain dataset. Following prior studies [9, 18], 10% of the 360K samples are for fine-tuning, while about 119K remain for evaluation. A linear classifier head is added to the encoders, trained with a multi-label soft margin loss. Alongside the foundation models from Tab. 1, we incorporate ConvNeXtV2 [28], which was trained explicitly for this downstream dataset by Kai *et al*. [3]. The fine-tuned models are evaluated using the mean Average Precision (mAP) metric across 19 classes.

Change Detection. The OSCD dataset [5] is employed, comprising 14 MS samples for training and 10 for evaluation. Each pair of images is simultaneously processed through a shared encoder, and their difference is then passed to UperNet. The models are fine-tuned for 60 epochs with a batch size of 32 and a learning rate of $1e-3$, and the final performance is measured using the F1 score.

Complete & Incomplete Dataset. For each downstream task, we evaluate the fine-tuned models using two types of datasets, *i.e.*, complete dataset and incomplete dataset. For the complete dataset, we retain all data in its original form. For the incomplete dataset, which is not publicly available, we randomly remove values from the complete datasets [22]. To simulate sensor degradation or extreme atmospheric conditions [17, 22], we randomly remove invisible spectral bands and time points from the sequence of both MS and SAR data. The visible bands, *i.e.*, RGB bands, are retained to preserve the essential information for model learning [2]. We implement three missing rates: 20%, 30%, and random ($Rand.$). The fixed rates remove 20% and 30% of invisible bands and time points, while the $Rand.$ setting applies a variable rate for each spectral band sequence and temporal sequence.

Table 2. Segmentation results (mIoU↑) for models fine-tuned on datasets with varying missing rates and diverse input types, ranging from static uni-modal to temporal (T-) multi-modal data. $Rand.$ indicates a random missing rate applied to each sequence.

Model	Input	Missing rate			
		0%	20%	30%	*Rand.*
SatlasNet [1]	MS	41.57	40.81	38.97	39.09
	T-MS	43.51	43.02	41.38	41.69
	SAR	36.22	–	–	–
SpectralGPT [9]	MS	42.67	42.22	41.31	41.21
	T-MS	45.69	44.85	43.15	42.49
Satmae++ [18]	MS	42.90	42.43	41.97	40.07
Croma [7]	MS	40.77	39.55	39.14	38.59
	SAR	36.48	–	–	–
	MS-SAR	44.01	43.80	42.97	41.45
RobSense (Ours)	MS	43.98	43.29	42.87	42.76
	T-MS	<u>46.11</u>	<u>45.15</u>	<u>43.93</u>	<u>43.88</u>
	SAR	38.33	–	–	–
	MS-SAR	44.94	44.43	43.62	42.96
	T-MS-SAR	**46.36**	**45.81**	**44.72**	**44.26**

4.2. Performance on Downstream tasks

Semantic Segmentation. Tab. 2 compares foundation models fine-tuned on datasets with missing rates from 0% to a random rate ($Rand.$). Most models show declining performance as the missing rate increases. However, SatlasNet improves from a 30% missing rate to $Rand.$

In comparison, RobSense consistently outperforms baselines in terms of mIoU across both missing rates. On the complete dataset using uni-modal data, RobSense achieves the highest mIoU of 46.11, surpassing the second-best model, SpectralGPT with sequential MS input (T-MS), by 0.92%. Fig. 3 illustrates that RobSense effectively captures seasonal changes in dynamic areas, such as shrublands and crop fields. In multi-modal settings, RobSense reaches a mIoU of 46.36, outperforming SpectralGPT by 1.47% and Croma by 5.34%. This highlights RobSense's ability to effectively capture time-specific distribution through the temporal distribution

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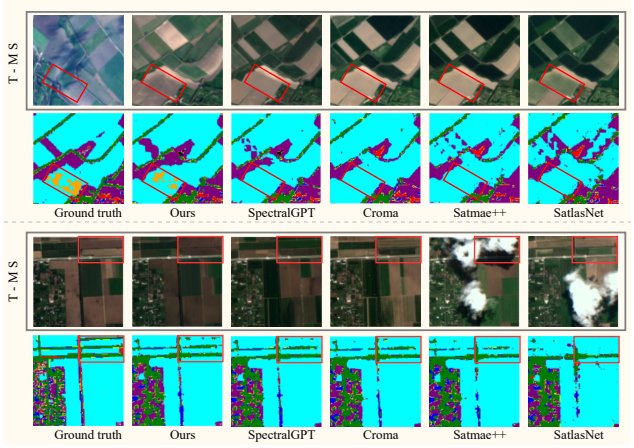


Figure 3. Visual comparison of fine-tuned models on the segmentation dataset. RobSense (ours), SpectralGPT, and SatlasNet receive sequential MS images, while Croma and Satmae++ utilize the static MS image with the best result.

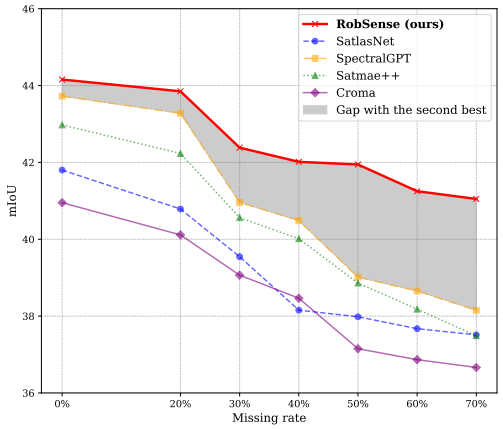


Figure 4. Performance comparison of foundation models fine-tuned on the segmentation dataset with static MS input under different missing rates. The performance gap relative to the second-best model, SpectralGPT, widens as the rate of missing data increases, underscoring the effectiveness of the proposed latent reconstructor in handling incomplete inputs.

alignment of multi-modal data utilizing multivariate KL divergence. At missing rates 20% and 30%, RobSense outperforms the two next best models, *i.e.*, SpectralGPT and Croma by 0.96-1.57% and 1.75%-2.01%, respectively. Under random missing rate, RobSense maintains the highest mIoU at 44.26. Specifically, Fig. 4 illustrates each model's performance at varying missing rates in the *Rand.* dataset. The proposed model consistently outperforms the others across all missing rates, with performance gaps widening as the missing rate increases. Notably, it demonstrates a significant advantage at a 70% missing rate, where only the RGB bands remain. This demonstrates RobSense has strong performance across missing rates by effectively employing latent reconstructors to reconstruct the rich representations from RGB information.

Classification. Tab. 3 compares foundation models fine-tuned on the BigEarthNet dataset at varying missing rates. Overall, all models show a performance decline as missing data increases across input types. The proposed model achieves the highest mAP across complete and incomplete datasets with multi-modal data.

On the complete dataset, Croma's performance declines when incorporating MS-SAR data compared to using MS data alone. This is likely due to misalignment in the patch-level multi-modal pairs, resulting in information loss in the multi-modal representations. Tab. 4 demonstrates that Croma fails to predict the multi-modal input with large and complex regions such as mixed forest or complex cultivation patterns. In contrast, ConvNeXtV2 achieves better results by effectively stacking MS-SAR data pairs. Additionally, while RobSense lags behind SpectralGPT and Croma with uni-modal MS input, it outperforms both models when SAR data is incorporated, improving by

Table 3. Classification results (mAP ↑) for models fine-tuned on datasets with varying missing rates. *Rand.* indicates a random missing rate applied to each sequence.

Model	**Input**	**Missing rate**			
		0%	20%	30%	*Rand.*
SatlasNet [1]	MS	82.8	80.83	79.76	76.21
	SAR	74.21	–	–	–
SpectralGPT [9]	MS	87.50	84.68	81.90	81.21
Satmae++ [18]	MS	85.11	83.65	80.79	78.91
Croma [7]	MS	87.58	84.86	82.96	82.69
	SAR	76.06	–	–	–
	MS-SAR	86.28	83.70	83.27	82.33
ConvNeXtV2 [28]	MS	84.89	80.61	76.56	73.12
	SAR	78.90	–	–	–
	MS-SAR	85.10	82.55	77.74	75.19
RobSense (Ours)	MS	86.36	85.28	84.36	83.69
	SAR	81.11	–	–	–
	MS-SAR	**88.17**	**87.23**	**85.46**	**84.87**

Table 4. Comparison of multi-label classification for RobSense and Croma, with multi-modal input. Correct predictions are marked with ✓ and incorrect predictions with ~~strikethrough text~~.

MS	**SAR**	**Ground truth**	**RobSense (ours)**	**Croma**
		Arable land	✓ Arable land	✓ Arable land
		Broad-leaved forest	✓ Broad-leaved forest	✓ Broad-leaved forest
		Coniferous forest	✓ Coniferous forest	✓ Coniferous forest
		Mixed forest	✓ Mixed forest	~~Pastures~~
		Arable land	✓ Arable land	✓ Arable land
		Complex cultivation patterns	✓ Complex cultivation patterns	~~Urban fabric~~
		Broad-leaved forest	✓ Broad-leaved forest	✓ Broad-leaved forest
		Coniferous forest	✓ Coniferous forest	✓ Coniferous forest
		Mixed forest	✓ Mixed forest	

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Table 5. Change detection results (F1↑) for foundation models fine-tuned on the MS-only dataset with varying missing rates.

Model	Missing rate			
	0%	20%	30%	*Rand.*
SpectralGPT [9]	54.29	54.04	52.23	50.09
Satmae++ [18]	56.28	54.34	54.10	53.20
RobSense (Ours)	**58.22**	**56.26**	**56.25**	**55.47**

Table 6. Impacts of Temporal Distribution Alignment (TDA) and Latent Reconstructors (Res) on model classification performance.

(a) Impact of TDA

Weights	mAP↑
Rand. Init	79.82
$\text{RobSense}_{\text{AE}}$	81.11
$\text{RobSense}_{\text{AE + Cons}}$	81.53
$\text{RobSense}_{\text{AE + TDA}}$	**84.19**

(b) Impact of Res

Weights	mAP↑
Rand. Init	73.13
$\text{RobSense}_{\text{w/o Res}}$	78.67
$\text{RobSense}_{\text{w/ Res}}$	**80.70**

0.77% and 0.67%, respectively. This suggests that the proposed model, pre-trained with multi-modal temporal distribution alignment, can effectively mitigate information loss and enrich multi-modal representations. On the incomplete dataset, as the missing rate increases to 20% and 30%, RobSense maintains its lead with mAPs of 87.23 and 85.46 in multi-modal settings, outperforming the second best models, *i.e*., Croma, by 2.37% and 2.5%, respectively. Under the random missing condition, RobSense achieves a strong mAP of 84.87, showing a margin of 2.54% over Croma and 3.66% over SpectralGPT.

Change Detection. Tab. 5 presents the quantitative results of SOTA models, *i.e*. SpectralGPT, Satmae++ for this task. Unlike the baselines, which either lack input diversity or suffer from information loss when data are incomplete, our model leverages TDA-based temporal multi-modal pre-training and reconstructed representations for robustness. Consequently, on the complete dataset, our model achieves an F1 score of 58.22, outperforming SatMAE++ and SpectralGPT by roughly 2–4%. Moreover, even under conditions of missing data (with 20–30% or randomly missing), our model maintains the highest performance, underscoring its resilience and effectiveness.

4.3. Ablation Study

We examine the impact of two key factors: temporal distribution alignment and latent reconstructors. Due to computational constraints and these factors being less influenced by data scale, the model is pre-trained on 5% of the dataset, with BigEarthNet used for fine-tuning on static multi-modal input. Tab. 6 presents the impact of the examined factors.

Temporal Distribution Alignment. To evaluate TDA's effectiveness, the model is pre-trained with three strategies: (i) Autoencoder loss (AE), (ii) AE + contrastive loss (AE + Cons), and (iii) AE + TDA. Meanwhile, other settings are kept the same. Starting from random initialization, the model's baseline mAP is 79.82. Pre-training with AE loss or AE + Cons loss leads to moderate improvements. Pre-training with AE or AE + Cons yields moderate improvements while incorporating TDA (AE + TDA) achieves the highest mAP of 84.19, indicating that TDA significantly enhances latent representations by aligning the distributions of both modalities consistently over time.

Latent Reconstructors. To evaluate the effectiveness of uni-modal latent reconstructors, we pre-train the proposed model under two settings: (i) with latent reconstructors (w/ Res), and (ii) without latent reconstructors (w/o Res). In the second setting, we add four additional ViT layers to the uni-modal encoders, maintaining a fair total parameter count. Additionally, for this setting, we pre-train with only one phase, *i.e*., temporal multi-modal learning, and apply three masking strategies during pre-training: patch, invisible band masking in MS images, and time-point masking with only one phase. Results show that both configurations outperform random initialization on incomplete datasets, but the model with reconstructors pre-trained by two phases achieves the highest mAP. In contrast, pre-training with one phase and all three masking strategies can result in significant information loss, thereby diminishing performance. These findings demonstrate that training latent reconstructors separately can effectively handle missing data by preserving rich representations.

5. Conclusion

In conclusion, we propose RobSense, a robust multi-modal foundation model for remote sensing that effectively integrates MS and SAR data. Our model addresses two key robustness challenges: supporting diverse input types and handling incomplete data. The model employs parallel uni-modal encoders and a multi-modal encoder to pre-train with temporal distribution alignment and a masked autoencoder strategy, enabling it to handle four input types flexibly. To manage incomplete data, RobSense includes two uni-modal latent reconstructors trained with masked spectral bands and time points, which effectively reconstruct rich representations even with missing data. Therefore, RobSense demonstrates strong performance on both complete and incomplete datasets. However, RobSense requires higher computational costs for pre-training due to its two pre-training phases. Future work will evaluate RobSense on a wider array of remote sensing tasks to assess its generalization in earth observation and to expand its capabilities to include language for tasks requiring both visual and linguistic understanding.

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